



CSE350: Digital Electronics and Pulse Techniques

Project Title: Digital Voltmeter (Without Arduino and IC)

Section: 06-A

Group: 2

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Abstract

To determine the voltage of an unknown source, we are using a 3-bit Flash ADC within the range of 0V to +7V. Firstly, in the Flash ADC, we are creating a resistor ladder using 8 resistors which is connected to ground on one side and to the +7V from the other. Our goal is to input the unknown voltage in the inverting terminal and the reference ladder to the non-inverting terminal of the Op Amp comparators to generate the active-low signals required by the Active-Low Priority Encoder. The output of the priority encoder is connected to NOT gate logic which determines the accurate output for the 7-segment-digital display to show the voltage of the unknown input.

Background

The basic portion of our project consists of Flash ADC. The Flash ADC is a very fast Analog to Digital converter that takes only a single clock for the conversion. In general, the fundamental components of the Flash ADC are Operational Amplifiers and Resistors.

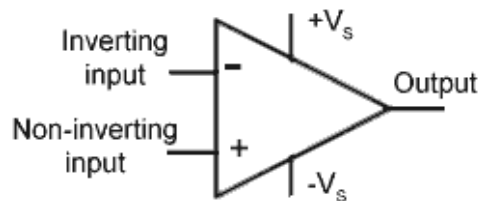


Figure: Operation Amplifier

An Op Amp comparator compares the voltage of two terminals and gives output voltage corresponding to its power rail, V_H or V_L . For example, if the input voltage of the inverting terminal is grounded (0V) and the non-inverting terminal's voltage is greater than the inverting input voltage and threshold voltage, the Op Amp saturates at V_H . The range of V_H and V_L generally varies from -8V to +8V.

For the input of the op-amp's non-inverting terminal, the voltage +7V is divided among the Op Amps (**LM324N**) using the Resistor Ladder. Using 8 resistors, in a series connection and taking the voltage within the interval of the resistors, voltage gets distributed among the Op

Amps to compare. With the 7 Op Amp comparators connected in parallel to receive sequential voltage difference, If the target element's voltage is inputted in all the Op Amps' inverting terminal, the Op Amps that have higher reference voltage than the target will output a HIGH voltage and feed to the 8x3 Active Low Priority Encoder (**SN74HC148**). According to Texas Instruments [1], the SN74HC148 ensures that only the highest order data line is encoded during the analog-to-digital conversion process. Since multiple comparators are triggered at the same time, this encoder specifically prioritizes the highest active voltage level to determine the final 3-bit binary output.



Figure: 8x3 Active-Low Priority Encoder

The Flash ADC follows this **Mid Rise Quantization Scheme**:

Quantization Interval	Quantization Level	Digital Output
[0-1)	0.5V	000
[1-2)	1.5V	001
[2-3)	2.5V	010
[3-4)	3.5V	011
[4-5)	4.5V	100
[5-6)	5.5V	101
[6-7]	6.5V	110

The binary output is then fed into the NOT gate for the active low BCD to 7-segment decoder. In our project, the NOT gate is constructed using a NAND (**74HC00**) gate. After inverting the logic outputs, it is fed into the BCD to 7-segment decoder (**74LS47**) for the output required to show digitally with the help of a **Common Anode 7-segment display**. As specified by Texas Instruments [1], the SN74LS47 features active-low outputs, making it ideal for directly driving the common-anode 7-segment display used in this digital voltmeter circuit.

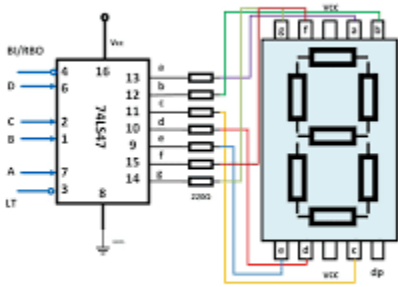
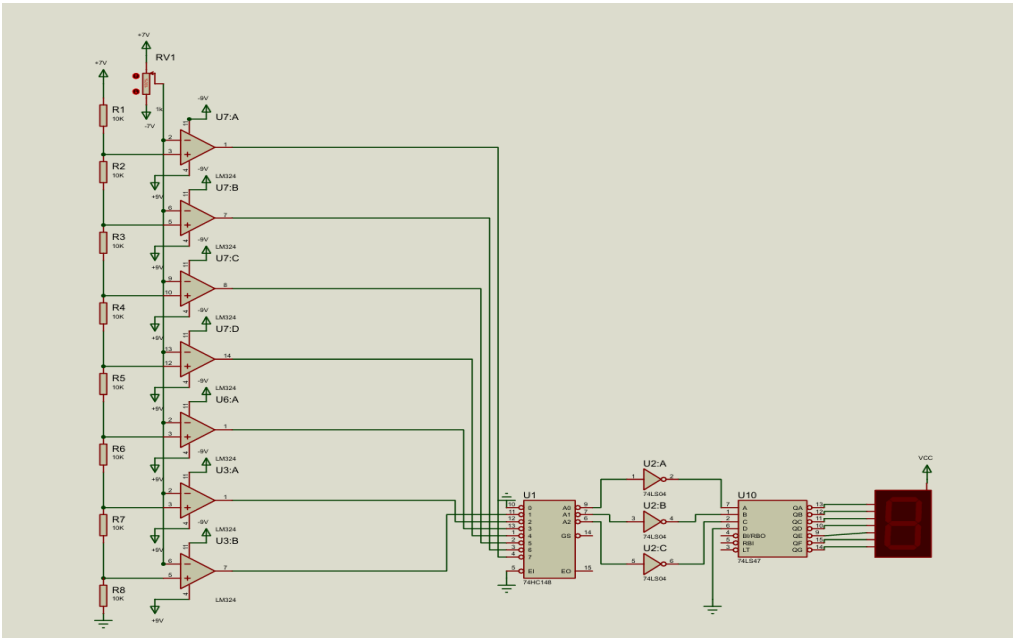
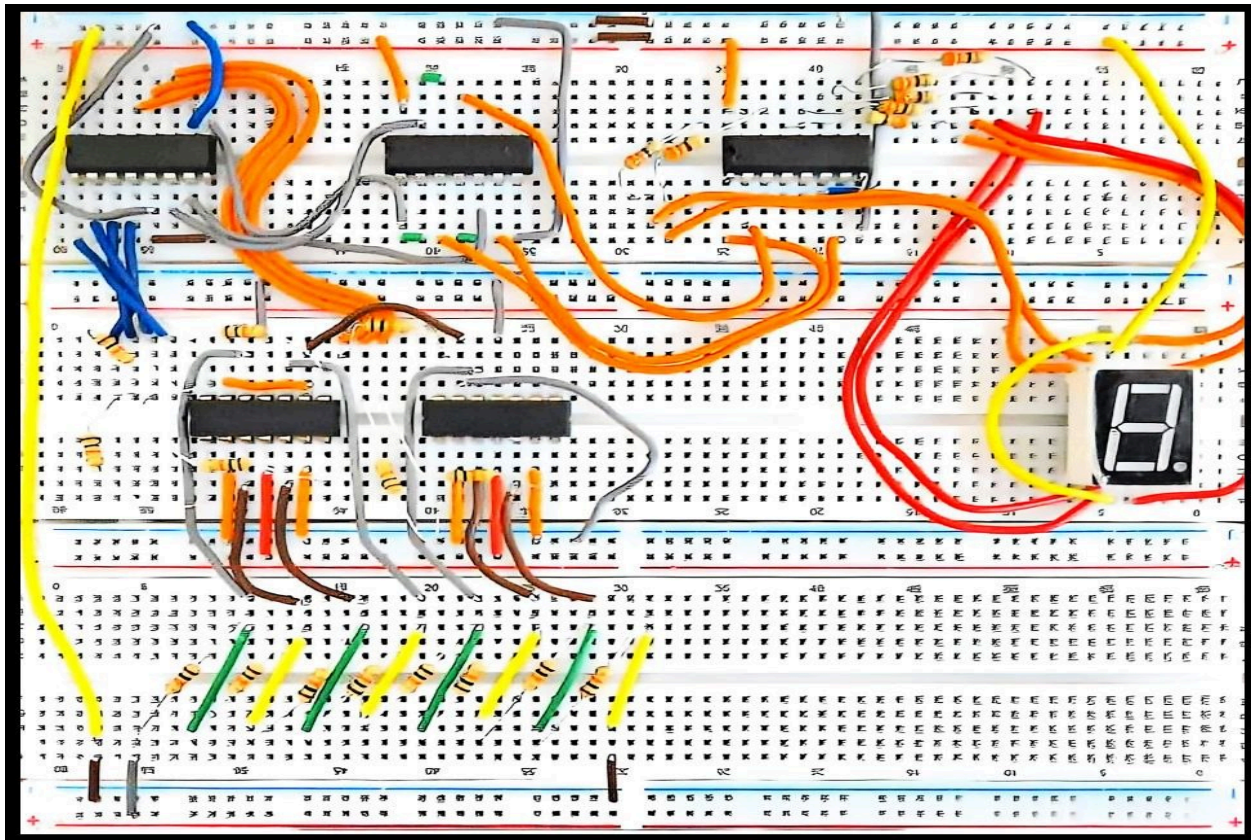


Figure: BCD to 7-segment decoder with 7-segment display.



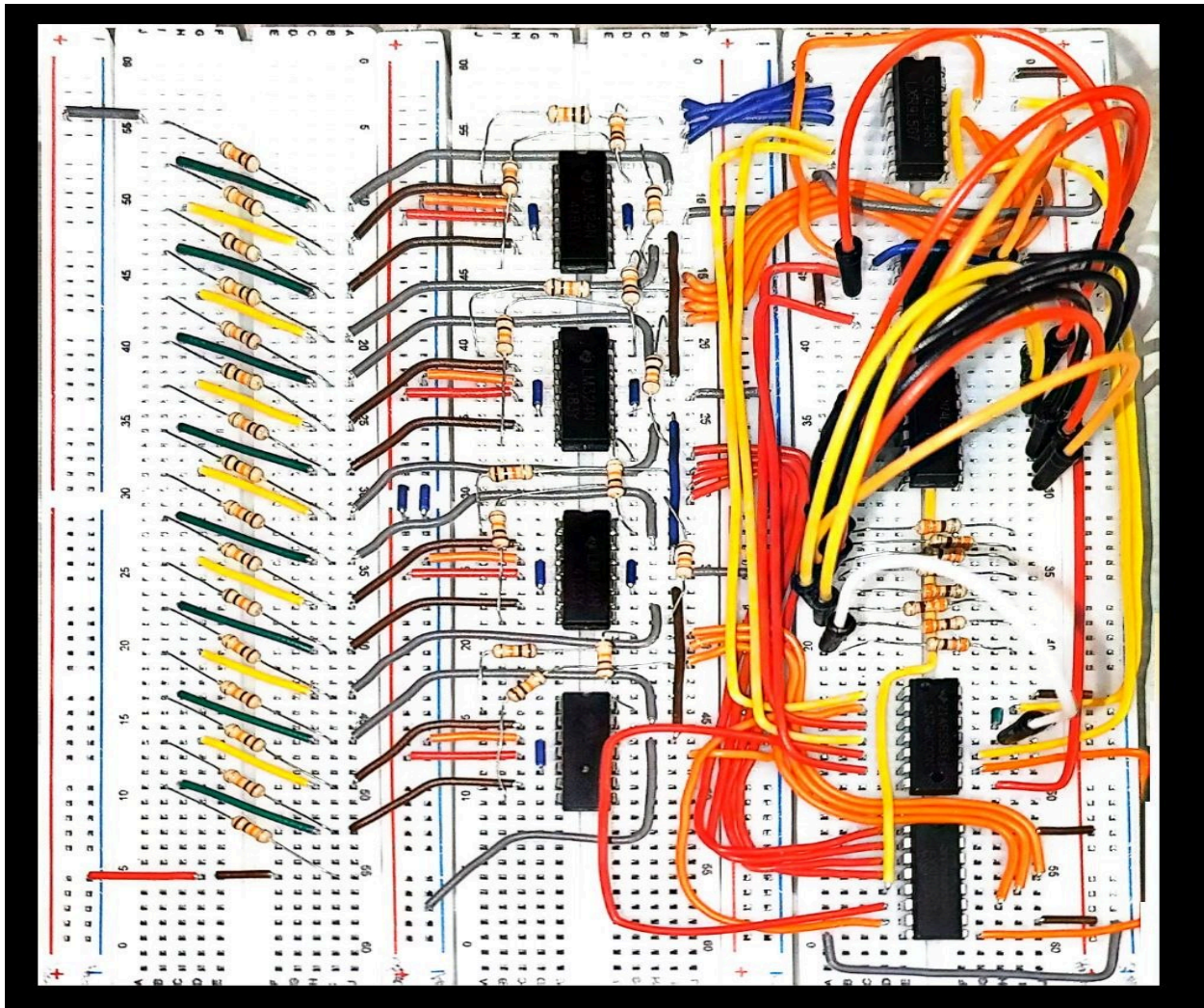
Below is the Simulation of the voltmeter using 3-bit Flash ADC ranging from 0V to +7V:



Challenges

We initially tried creating a 4-bit Flash ADC. However, implementing a 4-bit flash ADC voltmeter required an excessive amount of wires and components which has introduced heavy noise within the circuit board. While building the circuit, we struggled with the implementation of negative voltage. We came up with the method of shorting the positive and negative terminal of the voltage generator's two channels and using the positive and negative from two different channels as positive 7V and negative 7V. Since we were using different components with higher voltage, we kept the current capped at 0.3 Ampere. Even after implementing everything according to the simulation of our desired output and checking every component validity using a component checker as well as the resistor, we could not get an accurate display from our implemented circuit. Lastly, we came to the conclusion that a voltmeter with 4-bit Flash ADC is creating enough noise so that it is giving wrong outputs.

Below is the Simulation of the voltmeter using 4-bit Flash ADC ranging from -7V to +7V:



Summary:

For a voltmeter with 3-bit Flash ADC, there is less noise or any significant disturbance while showing the digital output. The tested data of the implemented circuit matches almost identically with the quantization level we have focused to achieve. In conclusion, We believe that with higher quality of Integrated circuits, and considering noise margin for the IC's, the project can be implemented for higher lengths of values for detection.

Reference

- [1] Texas Instruments. (2022, March). *SNx4HC148 8-line to 3-line priority encoders* (Rev. H) [Data sheet].
https://www.ti.com/lit/ds/symlink/sn74hc148.pdf?ts=1777913941644&ref_url=https%253A%252F%252Fwww.google.com%252F
- [2] Texas Instruments. (n.d.). *BCD-to-seven-segment decoders/drivers* [Data sheet].
<https://www.ti.com/lit/gpn/SN54LS47>
- [3] Texas Instruments. (2025, September). *LMx24, LMx24x, LMx24xx, LM2902, LM2902x, LM2902xx, LM2902xxx quadruple operational amplifiers* (Rev. SLOS066AE) [Data sheet].
<https://www.ti.com>
- [4] Texas Instruments. (2021, August). *SNx4HC00 quadruple 2-input NAND gates* (Rev. SCLS181H) [Data sheet]. <https://www.ti.com>